

Ökobaudat Material Matcher

AI-assisted EPD matching pipeline · Claude API + Ökobaudat REST API · Python / Google Colab

Context: This document explains the automated pipeline developed as part of the technical assessment. It shows how a building element input flows through four steps to produce an EPD-matched, standards-verified output. Results from this pipeline informed the GWP calculations in Q1 and Q3.

Full code with README: github.com/annrosem/NeoBIM-Assignment → [okobaudat_pipeline.ipynb](#)

SECTION 01

What the Pipeline Does

Given a building element description (name, type, total thickness), the pipeline automatically decomposes it into standard material layers, finds matching Environmental Product Declarations (EPDs) in the Ökobaudat database, and verifies the assembly against German construction standards — without any manual database lookup.

PIPELINE AT A GLANCE

Input	<i>Building element dict (name, type, thickness)</i>
AI model	Claude Opus claude-opus-4-6 · Anthropic API
EPD database	Ökobaudat OEKOBAUDAT 2021-II · public REST API
Output	<i>JSON + Markdown compliance report</i>
Runtime	~60–90 seconds end-to-end

SECTION 02

Step-by-Step Process

```
USER INPUT

element = {
  "name":      "External Wall",
  "elementType": "Wall",
  "thickness": 300 # mm
}
```



01

LAYER DECOMPOSITION

Standards-Based Material Layer Breakdown

Claude is given the element definition and instructed to use its built-in **web search tool** to look up the relevant German and EU construction norms. It then decomposes the element into individual material layers (inside → outside order) that conform to those norms.

Norms searched:

- DIN 4108-2 — minimum thermal insulation requirements
- DIN 4108-3 — vapour barrier / moisture protection
- DIN 18531 — flat roof waterproofing
- EN ISO 6946 — thermal resistance calculation method
- GEG 2024 — Gebäudeenergiegesetz (Building Energy Act)
- DIN 1045 — concrete and reinforced concrete structures

Output per layer: name, thickness_mm, description, standard_ref, search_keywords[]

EXAMPLE OUTPUT — 300MM EXTERNAL WALL

- Layer 1: Interior plaster 15 mm — DIN 18550
- Layer 2: Masonry (Mauerziegel) 240 mm — DIN 1053
- Layer 3: Mineral wool insulation 160 mm — DIN 4108-2
- Layer 4: Exterior render 20 mm — DIN 18550

02

EPD SEARCH

Ökobaudat REST API Query per Layer

For each layer, the pipeline queries the Ökobaudat public API by name. It tries the primary layer name first, then alternative **search_keywords** generated by Claude, then a single-word fallback — stopping as soon as results are found.

API endpoint:

- GET /datastocks/{uuid}/processes?search=true&name={query}&format=json
- Datastock: OEKOBAUDAT 2021-II — publicly accessible, no auth required

- Returns up to 10 candidate EPD records per query
- Deduplicates across keyword attempts by UUID

Bug fixed here: API returns `name` inconsistently — either a plain string *or* a localised list. Both cases handled explicitly.

BUG FIX — API INCONSISTENCY

The Ökobaumat API returns `name` as `"Interior plaster"` (string) for some records and `[{"lang": "en", "value": "..."}]` (list) for others. Calling `.get()` on a string character caused `'str' object has no attribute 'get'`. Fixed with an `isinstance()` guard.

EXAMPLE — CANDIDATES FOUND FOR "MASONRY"

```
uuid: 30514538-fcb4-483b-b5d5...
name: Mauerziegel (ungefüllt)
uuid: a1b2c3d4-...
name: Kalksandstein
... up to 10 candidates
```

03

EPD RANKING

Best EPD Match Selected by Claude as LCA Expert

For each layer, Claude is prompted as an LCA specialist. It receives the layer specification and all candidate EPDs, and selects the single best match — returning a UUID, name, confidence level, and a one-sentence reasoning.

Confidence levels:

- **HIGH** — material type, density, and application closely match
- **MEDIUM** — correct material family, minor differences
- **LOW** — only partial match available
- **NONE** — no relevant EPD found in database

Retry logic: up to 3 attempts with 1s backoff on API errors

EXAMPLE OUTPUT

```
layer: "Masonry (Mauerziegel)"
best_uuid: 30514538-fcb4-483b...
best_name: Mauerziegel (ungefüllt)
confidence: high
```

reasoning: "Unfilled clay brick at 575 kg/m³ matches
standard load-bearing masonry for this wall type."

04

STANDARDS VERIFICATION

Full Assembly Checked Against German Norms

The complete matched assembly is passed to Claude as a building physicist. It verifies the assembly layer-by-layer against the applicable standards, checks for missing mandatory layers, and issues an overall compliance verdict with a score out of 100.

Verdicts:

- COMPLIANT — all layers present, thicknesses and U-value meet norms
- PARTIALLY COMPLIANT — assembly meets most requirements with minor gaps
- NON-COMPLIANT — mandatory layer missing or critical requirement not met

Standards checked: DIN 4108-2, DIN 4108-3, DIN 18531, EN ISO 6946, GEG 2024



```
FINAL OUTPUT — OKOBAUDAT_RESULT.JSON + MARKDOWN REPORT

element · decomposition · matched_layers[ ]
  → layer · best_match { uuid, name, confidence, url }
  → candidates[ ]
verification { verdict · score/100 · layer_checks[ ] · recommendations[ ] }
report_markdown (full human-readable summary)
```

SECTION 03

How This Pipeline Feeds Q1 and Q3

QUESTION	ELEMENT	PIPELINE ROLE	MANUAL STEP STILL REQUIRED
Q1	External load-bearing wall, 435 mm	Layer decomposition + EPD UUID lookup per layer	GWP × quantity calculation in eLCA / spreadsheet
Q3	25 m² studio: floor slab + walls + flat roof	Layer decomposition + EPD candidates for all 3 elements	Area inputs + lifecycle stage selection in eLCA v1.0.1
Both	—	Standards verification confirms assembly is code-compliant before	Reviewer sign-off on AI-generated compliance

QUESTION	ELEMENT	PIPELINE ROLE	MANUAL STEP STILL REQUIRED
	LCA		assessment

In a production BIM-LCA workflow, element quantities would be extracted automatically from the IFC model, and the pipeline output would feed directly into an LCA calculation engine — eliminating all manual steps above.

SECTION 04

Assumptions & Limitations

- AI compliance assessment is a first-pass review aid — not a substitute for a certified building physicist
- Datastock fixed to OEKOBAUDAT 2021-II; newer datastocks may have different UUIDs and EPD coverage
- EPD search is keyword-based — the Ökobaudat API does not support semantic search
- Layer thickness is distributed proportionally to the given total; norm minimums are respected first
- Ökobaudat API is public and unauthenticated, but may be rate-limited or blocked from cloud IP ranges
- Rate limit (30k tokens/min on free Anthropic tier) can interrupt Step 1 if web search is token-heavy
- Module D (beyond system boundary) and B6 operational energy excluded from scope throughout

SECTION 05

How to Run

CELL	ACTION	NOTE
1	Install dependencies	<code>pip install anthropic requests rich</code>
2	Load API key	From Colab Secrets — key never hardcoded in notebook
3	Define pipeline functions	No execution yet — change <code>MODEL</code> here if needed
4	Run pipeline	Edit element dict, run cell, JSON auto-downloads
Output	<code>okobaudat_result.json</code>	Full structured result including <code>report_markdown</code>

Ökobaudat Material Matcher · Pipeline Documentation Claude Opus 4 · OEKOBAUDAT 2021-II · Python 3.10+ ·
· Appendix Google Colab